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Utilization of AI/ML Technologies for Fuel Thermal-Hydraulic Methodology Development in Westinghouse

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Abstract — The nuclear industry has fully embraced the development of accelerated fuel qualification (AFQ) approaches to speed up the assessment and validation of new fuel designs with respect to performance and safety metrics. To support the AFQ approach to shortening the time to develop and qualify new fuel for higher plant performance, Westinghouse utilizes advanced modeling and simulation technologies as part of their integrated and comprehensive AFQ vision through improved fuel performance prediction under various operating conditions and accident scenarios.

This paper provides example applications, prioritized in Westinghouse using machine learning technology, for fuel thermal-hydraulic applications with methodologies that are under development for the prediction of critical heat flux for pressurized water reactor (PWR) fuel thermal margin assessment and surrogate model development for crud-induced power shift risk prediction to enhance PWR fuel operation performance.

Keywords — Thermal hydraulics, machine learning, nuclear fuel, methodology development.

Note — Some figures may be in color only in the electronic version.

I. INTRODUCTION

Qualifying new fuel designs has typically involved a lengthy process of design, testing, irradiation, data gathering, validation, analysis, and licensing, including multiple iterations through these various phases. Westinghouse is currently developing its vision for an integrated and comprehensive accelerated fuel qualification (AFQ) approach. The approach includes, on a high level, lower length-scale modeling, accelerated irradiation, separate effect tests, sensor development and pool-side measurements. Advanced modeling and simulation (M&S) approaches and methodologies are being developed to complement and support various aspects of AFQ

in predicting fuel performance in normal reactor operation and accident scenarios.

This approach will be leveraged to focus and accelerate data gathering, reduce iterations through the development, validation, and analysis processes, and provide a more robust licensing case. Furthermore, this method has the potential to reduce the time and costs associated with licensing and the implementation analysis of new fuel products. Part of the advanced M&S approach includes the utilization of artificial intelligence (AI)/machine learning (ML) technologies, which are finding a continuously increased number of applications and adoption in the nuclear power industry to reduce design, engineering, manufacturing, and operational costs and timelines, which are directly synergistic with the goals of AFQ.

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Westinghouse utilizes advanced AI/ML technology in many applications; some examples include loss-of-coolant accident safety margin predictions,^[1] core monitoring, fuel management optimization, manufacturing efficiency improvements, design optimization, and fuel inspections, to name a few. Westinghouse has also recently launched an industry pioneering nuclear-specific AI system called Hive,TM which includes a generative AI model called bertha.TM

This paper focuses on the use of AI/ML for thermal-hydraulic applications in Westinghouse that support the overall AFQ vision. Two independent Westinghouse prioritized projects are provided that make use of AI/ML technologies, including methodologies under development for the prediction of critical heat flux (CHF) for pressurized water reactor (PWR) fuel thermal margin assessment and surrogate model development for crud-induced power shift (CIPS) risk prediction to enhance PWR fuel operational performance.

II. CHF PREDICTIVE MODEL DEVELOPMENT USING ML TECHNOLOGY

II.A. Introduction

For PWR designs, one of the specified acceptable fuel design limits is to prevent overheating of any fuel rod in the reactor core due to reaching departure from nucleate boiling (DNB). DNB occurs when a fuel rod clad surface is overheated due to the formation of a local vapor layer, causing a dramatic reduction in heat transfer capability and leading to fuel rod failure and the release of fission products. In current plant analyses, margin to DNB is quantified as the departure from nucleate boiling ratio (DNBR), which is a ratio of predicted CHF to actual heat flux for a given channel and axial location within a fuel assembly.

Most CHF correlations, also referred to as DNB correlations, were developed from rod bundle test data based on local fluid conditions. Local fluid parameters, such as local equilibrium quality and local mass flow rate, are calculated using a subchannel thermal-hydraulic code at each axial node of subchannels. However, recent CHF tests have indicated that at different testing conditions, such as high bundle inlet temperature or low subcooling, CHF due to dryout can be significantly lower than the DNB heat flux at similar local fluid conditions.

A new approach is under development at Westinghouse utilizing ML technology for CHF predictions from a combined DNB and dryout test database.

The ML approach allows for a more extensive search of the proper model that predicts CHF well and is suitable for quantifying margins to DNB or dryout through DNBR in PWR design applications. The CHF predictive model is not only trained and validated with experimental data, but is also evaluated through sensitivity studies as compared to known DNBR behavior in the application domain. The ML-based model is being evaluated and incorporated into subchannel code VIPRE-W for potential PWR core design applications.

It is important to note that there are a significant number of studies in the literature utilizing ML technology for CHF predictions, both early studies^[2] and more recent studies.^[3–9] Although these studies describe various approaches to CHF predictions, CHF on PWR fuel rods is highly dependent on the fuel design characteristics (e.g., fuel geometry and mixing vane grid spacers), and thus, the ML model form ultimately depends on the test data utilized (in this case, Westinghouse design fuel bundle test data).

In addition, the correlations not only need to be able to predict test data accurately (which is usually the more straightforward step), but they also need to perform well in application/design space with physics-based behavior (e.g., DNBR decreases with power increase), a step often omitted from data prediction-only studies. Thus, the study presented herein adopts a unique approach with regard to the ML model details and the applicable design space.

The predicted CHF for DNBR relies on data collected from small-scale rod bundle tests typically in a 5×5 configuration simulating the fuel design reaching DNB. The local fluid conditions at DNB locations are predicted by a subchannel thermal-hydraulic code. Most DNB correlations developed for PWR design applications utilize the local fluid parameter approach linked with a subchannel code,^[10] as DNB is typically associated with local vapor formation on the fuel rod surface under subcooled boiling conditions.

Over the years, Westinghouse has developed and applied several DNB correlations (e.g., WRB-1^[11]) with the Westinghouse version of the VIPRE-01 code VIPRE-W^[12] for different PWR fuel designs. However, it is increasingly common for the 5×5 CHF rod bundle tests to encounter dryout of the liquid film in an annular flow regime under high-temperature or low subcooling conditions. This has been observed with a higher frequency for fuel designs containing thermal performance enhancement features, such as mixing vane (MV) grid spacers and intermediate flow mixer (IFM) grid spacers. If the database contains a significant amount of dryout data, it is challenging to develop

a correlation that can accurately predict both DNB and dryout phenomena using the traditional empirical correlation approach based solely on local fluid conditions. To address this challenge, a new approach using ML technology^[13] has been applied to generate a CHF predictive model based on a combined DNB and dryout database following the Westinghouse-established process for CHF testing, modeling, design code, and plant design applications.

II.B. DNB Correlation Development Process

The established process for CHF or DNB correlation development is illustrated in Fig. 1. The process starts with the design of test bundles that simulate fuel assembly characteristics important to DNB, with rod bundles typically limited to a small scale, typically in a 5×5 rod array.^[14,15] The test rods are electrically heated with thermocouples embedded within the rods at locations where DNB is most likely to occur. The DNB test data consist of bundle average power, inlet flow rate, and inlet temperature and pressure, in addition to the bundle power profile and geometric information.

After the CHF testing is completed, the test data are evaluated using a subchannel code to obtain the local fluid conditions that were present within each test, including equilibrium quality and mass flow in the hot channels. The DNB test data and the corresponding subchannel local fluid conditions are then utilized as input to predictive model development. The resultant model is subject to statistical evaluation of measured-to-predicted (M/P) CHF ratios for accuracy, data poolability, bias/trend check, and 95/95 DNBR limit determination.^[16] A sensitivity study is performed at plant DNB limiting conditions to check model suitability for design applications. A DNB predictive model or correlation is subject to regulatory approval prior to any plant design application.

Based on the local fluid parameter values obtained from the subchannel code, a DNB correlation (e.g., WRB-1^[11]) can typically be developed based on a linear relationship between CHF and local equilibrium quality,

$$q''_{CHF} = A - BX_{loc} \quad , \quad (1)$$

where q''_{CHF} is the predicted CHF, A and B are the coefficients containing the local flow rate and other fluid/geometric parameters, and X_{loc} is the local equilibrium quality.

The relationship in Eq. (1) is consistent with the changes between q''_{CHF} and the bundle inlet enthalpy observed in CHF testing. Because of the geometric scaling difference between the test bundles and a reactor core, the bundle inlet enthalpy is not typically used in a DNB correlation for PWR design applications. A DNB correlation is intricately linked to a subchannel code for its development and application because its CHF predictions depend on local fluid conditions obtained from the subchannel code.

The accuracy of DNB correlations based on the linear relationship between CHF and local quality is being challenged with recent test data collected at low subcooling/high inlet temperature conditions. This is because the CHF observed at high inlet temperature operating conditions could be significantly lower than that at lower inlet temperatures, even if the local quality values at CHF locations are similar. An evaluation of the test data has indicated that CHF signals from thermocouple traces have reflected dryout occurrences at high inlet temperature testing conditions, with the resultant local quality above 20%.

Liquid film dryout occurs in annular flow regimes with significant separation of liquid and vapor phases. Subchannel codes, such as VIPRE-01 or VIPRE-W, which are based on a homogeneous two-phase flow

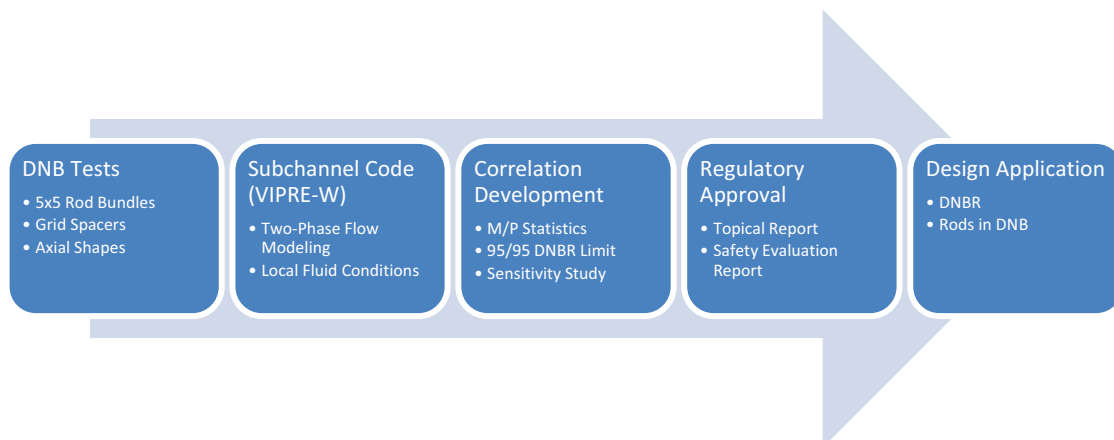


Fig. 1. DNB correlation development process.

model, may predict higher local quality associated with the dryout data and low subcooling if the effect of liquid and vapor separation in the annular flow regime is not considered. It is a time-consuming effort and intensive process to modify, validate, and license a subchannel code for reactor design applications. As the DNB correlation relies on the local fluid conditions from the subchannel code for input, a modification to the existing DNB correlation formulation is needed for more accurate predictions of both DNB and dryout test data if no change in the subchannel code or its two-phase flow modeling is made.

II.C. CHF Predictive Model Development Through ML

Current ML model development for CHF prediction is based on the existing correlation development process. There is no change in the process as shown in Fig. 1, including interface with the subchannel code. The changes are limited to the replacement of the empirical correlation, shown in Eq. (1), with a ML predictive model. The ML approach allows for the exploration of various model types and input parameters. A rigorous procedure has been established for validating the ML model and sensitivity study for potential design application.

II.C.1. Database Selection

The database for the initial ML model development and validation was based on data from rod bundle tests

applicable to Westinghouse PWR 17×17 fuel designs; an illustration of the 5×5 CHF test bundles used to simulate these designs are shown in Fig. 2. In all, this database contains approximately 2400 data points defined by the following:

1. Bundles containing fuel rods with outside diameters of both 0.360 in. (9.14 mm) and 0.374 in. (9.50 mm).
2. Bundles containing an unheated tube in the center to simulate the guide thimble tube of the fuel assembly.
3. Bundles with heated lengths ranging from 8 to 14 ft.
4. Axial spans within each bundle containing both MV grid spacers and IFM grid spacers.
5. Test operating conditions spanning a wide range of system pressures, inlet mass flux and inlet temperature, and power distribution defined by both uniform and cosine axial power profiles.

The database was also defined to contain both DNB and dryout data. Based on the thermocouple traces at the CHF locations obtained from the test reports, dryout data and their occurring conditions associated with low inlet subcooling could be evaluated and identified. Slightly more than 20% of the 2400 points could be considered as dryout data with low inlet subcooling (i.e., high inlet temperature and/or high inlet temperature with low inlet flow at a given pressure); these datapoints are typically associated with the local equilibrium quality above 20% at the CHF elevations.

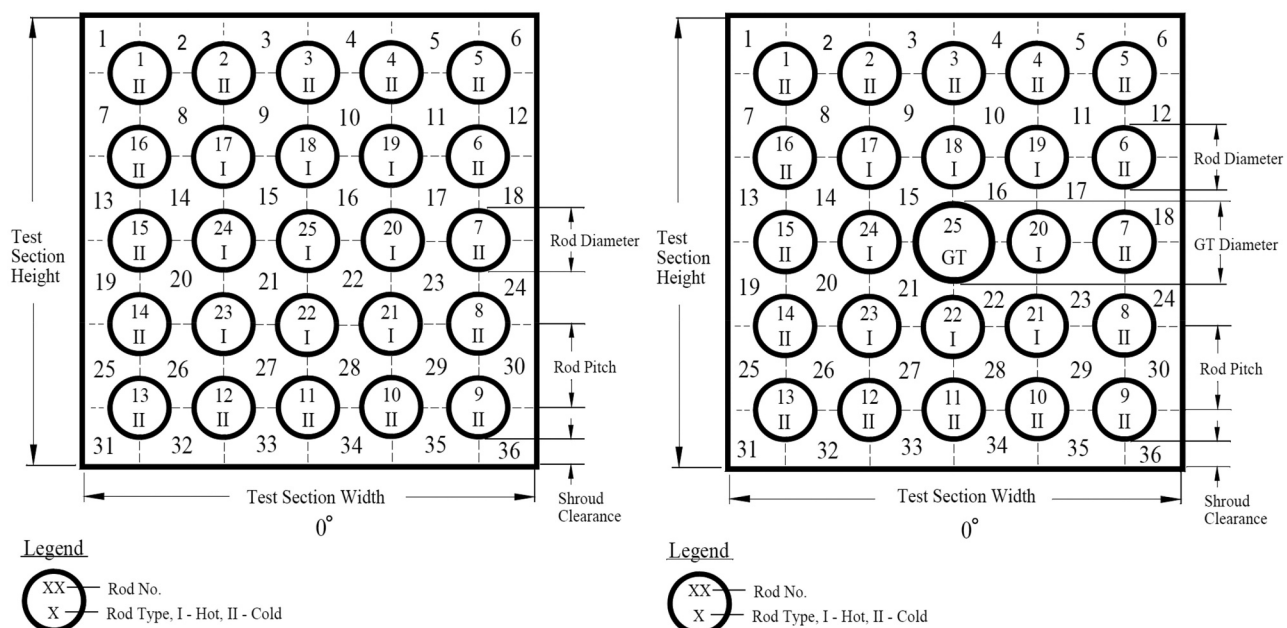


Fig. 2. Radial diagrams of 5×5 CHF test bundles.

II.C.2. Input Parameters to ML Modeling Approach

Most of the base input parameters utilized for ML correlation development remain unchanged compared to what has been utilized historically. These base inputs, which were obtained from the bundle test data and sub-channel code output, generally can be categorized as follows:

Coolant condition–related parameters, including system pressure defined by each test and local mass flux/local quality defined by the subchannel code.

Axial location–related parameters, including boiling length, heated length, and a F-factor, also referred to as the Tong factor^[17] to account for effects of boiling and nonuniform axial power distribution, as well as distance parameters which describe elevations relative to the upstream MV or IFM grid spacer.

Geometry-related parameters, such as rod diameter and rod-to-rod pitch, which describe characteristics related to a specific fuel design.

A notable addition to the base set of model inputs is the I_2 parameter, which is typically utilized in CHF correlations for boiling water reactor fuel.^[18] This term was included in an attempt to better account for dryout in the annular flow regime. The I_2 parameter associated with the boiling length is defined as

$$I_2 = \frac{1}{Lx(z_{d0})} \int_{z_0}^z x(z') dz', \quad (2)$$

where

L = bundle length

x = steam quality

Z_{d0} = axial position of dryout

Z_0 = axial position of boiling boundary.

While the base inputs themselves are largely consistent between traditional and ML-based correlations, it is also noted that the ML development efforts opted to manually combine several inputs through feature engineering before passing them to the model. This was done in an effort to provide the model with some amount of domain knowledge during training (e.g., CHF decreases as a function of distance from the previous mixing vane grid).

II.C.3. Model Development

The ML model development approach to predict CHF based on bundle test data was first described in

Ref. [13]. While several ML-based algorithms were found to be capable of generating good fits to CHF test data, it was identified that two key options available in the design of a neural network could help improve the generalization of predictions across the application space relative to other ML algorithms trained using the same input features.

First, the use of varying dropout rates for specific inputs and/or for specific layers across the model was found to aid in the prevention of overfitting to the training/validation space. This technique was observed to be particularly useful for axial location–based inputs, which provide valuable information (e.g., coolant mixing relative to grid locations and energy addition relative to the total heated length), but have a high propensity for overfitting due to low variance within the training/validation data (i.e., CHF tends to occur in similar axial locations across all empirical tests).

Second, it was determined that the translation of geometric characteristics into a multidimensional representation space using an embedding layer could also provide improvements to the model's predictive capability. At a high level, embeddings can be described as vectors that characterize real-world objects (typically words, images, or videos, but in this case fuel assembly geometries). While geometric characteristics are traditionally defined as continuous inputs to CHF correlations, defining them with embeddings of each unique fuel design allows the model to disregard intermediate geometries for which there is no physical representation (e.g., bundles with rod diameters of 0.365 in.). And given that the purpose of embeddings is to characterize fuel bundle geometry as a vector, they also allow the model to learn relationships between different geometries as it is being trained.

All neural network–based CHF models generated to date have utilized a sequential, fully connected architecture. This choice was made largely to facilitate the easy extraction of all weights and biases defined by a trained model that could then be directly coded into the FORTRAN-based VIPRE-W thermal-hydraulic code. For hyperparameter tuning, a Bayesian optimization approach was implemented to identify model attributes that resulted in the overall best tuning score. This score was defined to consider model performance not only in terms of the root-mean-square error for empirically measured versus correlation-predicted CHF, but also in terms of correlation performance within the application space defined by a set of typical inlet temperatures, system pressures, and core power levels.

Note that the mathematical formula describing how each of these terms were weighted within the tuning

score is considered to be proprietary information, and therefore is not published as part of this work. With a Bayesian approach, performance of the model with previous choices for hyperparameters can be leveraged to inform what hyperparameters should be evaluated next, which provides an advantage relative to other brute force approaches to hyperparameter tuning (e.g., random or grid-based selection of hyperparameters).

II.C.4. Model Validation and Uncertainty Study

Consistent with the validation of traditional DNB correlations, ML-based correlation performance was validated with the test data according to statistics describing the M/P CHF ratio (the reciprocal value of the DNBR). As highlighted in Table I and illustrated in Fig. 3, the neural network-based correlation provides a good fit to empirical test data with no significant trends/biases to either DNB or dryout. Most of the high-temperature or low subcooling data identified in Fig. 3 were observed as representative dryout data during the tests, which simulated the 17×17 fuel design with a fuel rod outside diameter of 0.36 in. (9.1 mm) and IFM grid spacers. The following points were also noted:

1. Additional checks of M/P distributions with respect to bundle pressure, channel local mass flux, channel local quality, modeling parameters (e.g., the I_2 parameter), and test bundles (e.g., typical and thimble bundles) at

TABLE I

CHF M/P Statistics for Combined Training/Validation Data Sets

CHF M/P mean	1.002
M/P standard deviation	0.052
Number of data points	2405
Estimated 95/95 limit	1.14

different subregions of the fluid conditions showed no significant trend or bias in the ML CHF predictions.

2. The ML-based correlation was also found to provide good estimates of model-predicted minimum DNBR locations relative to measured elevations in the empirical tests, and reasonable sensitivities in DNBR as a function of axial location (i.e., decreasing DNBR with increasing distance from the previous grid).

3. The 95/95 limit was estimated using Owen's factor for one-sided tolerance limits^[19] for the entire database containing both training and validation sets. It was also evaluated with different subgroups in the database as part of a subgroup poolability evaluation.^[20]

The ML-based model also took steps to expand on the traditional use of M/P statistics for the model validation described previously. In particular, the definition of allowable inputs to the model was enhanced through the definition and application of an out-of-distribution detector. This methodology acts to restrict inputs based on

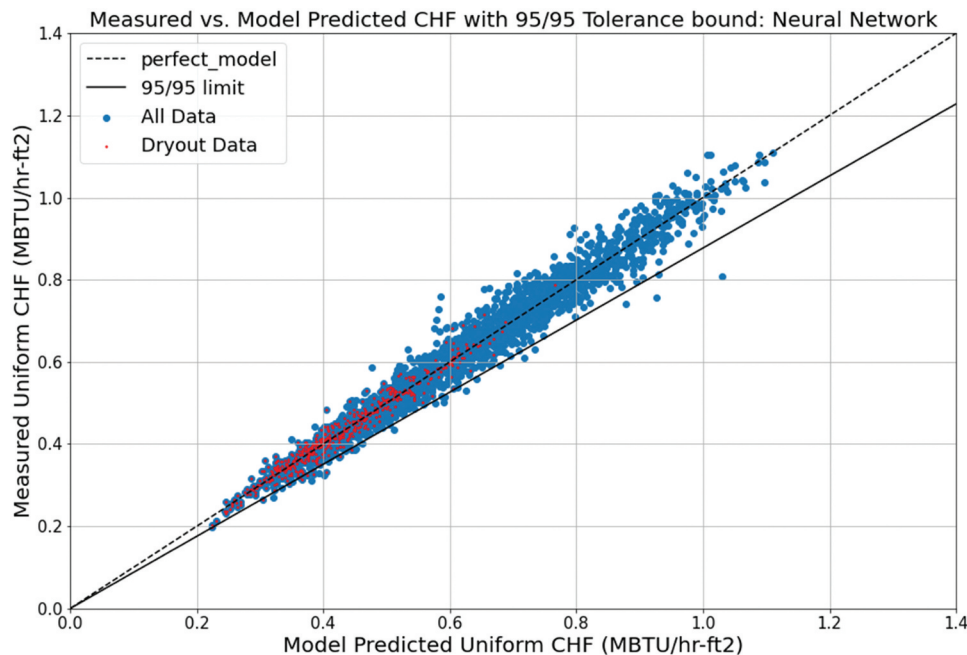


Fig. 3. Measured CHF versus predicted CHF from the ML model.

their distance from the training data and prevents the model from being applied in regions that may have high uncertainty.

Additionally, a custom loss function was utilized to enable the prediction of not only a best-estimate CHF, but also a conservative 95% lower bound value. The lower bound prediction interval learned during training was transformed into a rigorous form of uncertainty with marginal coverage guarantees through calibration with a conformal prediction process. The benefits of a calibrated lower bound prediction interval are twofold in that (1) comparison between lower bound and best-estimate predictions of CHF provide a method for the quantification of predictive uncertainties that can be compared to the traditionally calculated 95/95 DNBR limit pooled over all data, and (2) calculated uncertainties can be utilized to identify potential changes in model accuracy across a high-dimensional application space.

II.C.5. Model Sensitivity Study with Subchannel Code

As part of the development process, sensitivity studies were performed using the ML DNB model linked with VIPRE-W on the predicted CHF behavior in the application domain. Unlike the test bundles in the data domain, the VIPRE-W model for the application domain consisted of subchannels and fuel assemblies in the reactor core, as illustrated in Fig. 4.

The effects of flow and enthalpy distributions on DNBR calculations are more significant in a reactor core model than in the test bundle model. As the M/P or DNBR should be close to unity in the data domain, the DNBR values in the application domain are typically

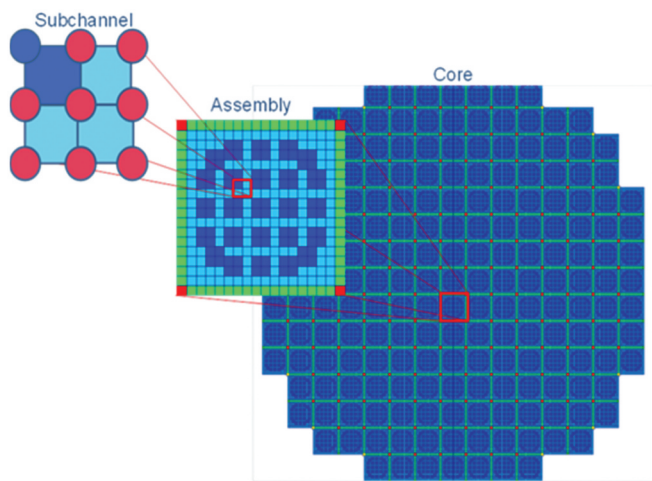


Fig. 4. PWR Core, fuel assembly, and subchannel diagram.

greater than the 95/95 DNBR limit. In addition to the DNBR limit, the application domain is restricted by other design constraints associated with DNBR predictions, such as the maximum quality threshold in the hot channel. In the application domain, the DNBR is expected to increase with respect to power decrease, following an overall trend with a negative slope.

As illustrated in Fig. 5, the DNBR predictions from the ML model linked to VIPRE-W confirmed appropriate model sensitivities (in this case, decreasing DNBR with increasing core average linear heat rate) at different core design conditions in the application domain. The core design conditions were obtained from DNB limiting conditions, such as the core thermal limits comprised of different power, pressure, and temperature, and low flow conditions from flow-reduction accidents.

Different radial and axial power distributions were also considered in the evaluation of the ML DNB model. In addition to the power, additional sensitivity studies were performed on the predicted DNBR to other core design parameters, such as inlet temperature and flow rate. The sensitivity study was to check and confirm the appropriateness of the ML DNB model for the reactor design applications.

II.D. Summary of ML Model Development for CHF Predictions

Recent CHF tests have indicated that at different testing conditions, such as high bundle inlet temperature or low subcooling, CHF due to dryout can be significantly lower than the DNB heat flux at similar local fluid conditions obtained from a subchannel code based on the homogeneous two-phase flow model. A new approach is under development at Westinghouse utilizing ML technology for selecting and evaluating flow channel and fluid parameters and their impacts on CHF predictions from a combined DNB and dryout database without any change in the DNB prediction process or the subchannel code.

The ML approach allows for a more extensive search of the proper model that predicts CHF well and is suitable for quantifying margins to DNB through the DNBR in PWR design applications. Different ML modeling approaches using neural network methods and parameter combinations were explored to determine a predictive model. The CHF predictive model was trained and validated with experimental data for solution accuracy. The ML-based model has been successfully linked to subchannel code VIPRE-W for testing and evaluation. The

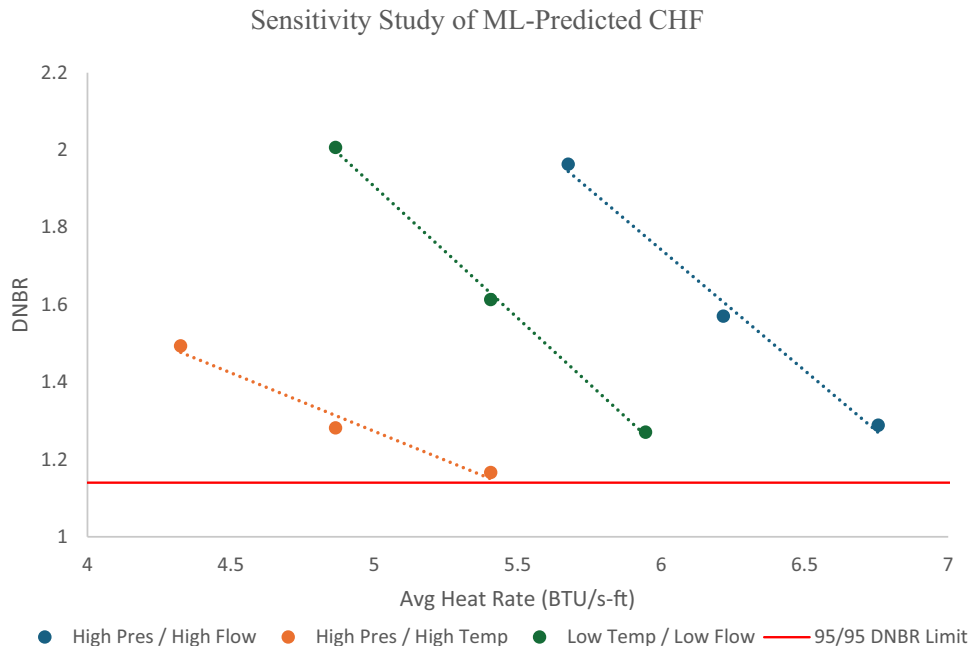


Fig. 5. DNBR versus average linear heat rate in the design domain.

ML DNB correlation has been rigorously tested and evaluated for applications in design space, with the performance compared to past licensed DNB correlations by Westinghouse.

The evaluations also include verification, validation, and uncertainty quantification of the ML model through the expected application domain. The initial ML model for the DNBR margin predictions showed significant improvements over the correlation based on the traditional approach. The model combined with VIPRE-W will be subject to additional evaluation in accordance with the credibility assessment guidelines for regulatory review and approval prior to reactor design applications.^[21]

The development and AFQ of new fuel requires an accurate prediction of the DNB limit as a critical safety margin. Typical/empirical DNB correlation development times will be drastically reduced (by more than a year) through the development of a DNB methodology using ML. It is important to note that the time savings also translate into cost savings, which supports initiatives on new fuel development.

Other benefits that apply to AFQ include better performance in the design/application space, in addition to potentially capturing/predicting DNB behavior through fewer experimental tests. This is especially important for future initiatives for physics-informed DNB ML modeling using high-fidelity computational fluid dynamics simulation data. The robust ML methodology can also

potentially be used for DNB correlation development for other reactor types in the future (which is being explored by Westinghouse).

III. IMPROVEMENT OF CRUD PREDICTION USING ML TECHNOLOGY

III.A. Introduction

Another application of ML technology within the area of thermal hydraulics at Westinghouse is improvement in crud risk predictions.^[22] Crud deposition on fuel cladding is a result of coolant chemistry and the corrosion of primary system components during the normal operation of a nuclear reactor. This material, which accumulates on the upper spans of high-duty fuel that has undergone subcooled nucleate boiling, may lead to decremented reactor performance in some cases because the deposition of crud acts as an additional barrier to fuel rod heat transfer.

Crud deposits can also affect both reactor power peaking and fuel element cladding integrity through CIPS (Fig. 6) and crud-induced local corrosion (CILC) (Fig. 7) phenomena, respectively. CIPS, which is the result of the incorporation of boron within crud deposits, alters power distributions by producing a significant negative axial offset (AO) deviation or shifting power from the upper to the lower elevation spans. A CIPS event erodes a plant's

shutdown margin as well as operational flexibility by the operator, and in more severe cases, requires power reduction for the remainder of the cycle to gain the eroded shutdown margin. CILC accelerates the corrosion of cladding material as a result of thick crud deposits that restrict flow and induce sudden dryout at the clad and crud interface. Corrosion affects the cladding integrity of the fuel rods, and in a worst-case scenario, can result in a breach of the cladding and the release of fission products into the coolant.

The CIPS/CILC phenomena are typically assessed using the Boron-Induced Offset Anomaly (BOA) computer code.^[23] BOA is utilized as a part of the standard engineering process in Westinghouse to provide both a characterization of the distribution of crud within a reactor core and the risk for occurrence of CIPS/CILC within a given cycle. The process of the calculation is driven by fuel management, with the core neutronics providing the core heat flux distributions via the ANC (Advanced Nodal Code, which is a multidimensional nodal code for all nuclear core design calculations) core model. The FUELDUTYDRV code can buffer the neutronics model and thermal-hydraulic conditions via the VIPRE-W code to provide the inputs necessary for the crud calculations completed by the BOA code.

Predictions from the BOA code are judged to need further improvement on automation efficiency with existing vendor code packages and improved characterization of the available margin to CIPS. For example, the

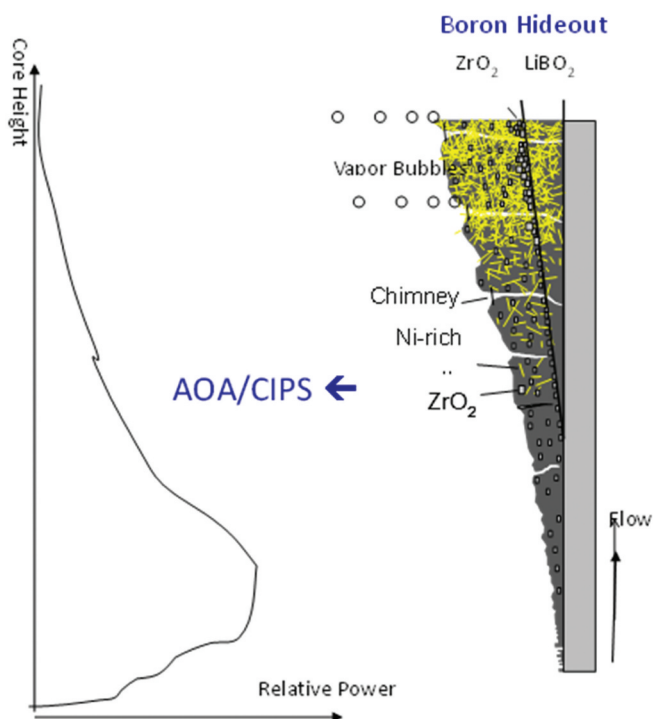


Fig. 6. Illustration of crud and CIPS.

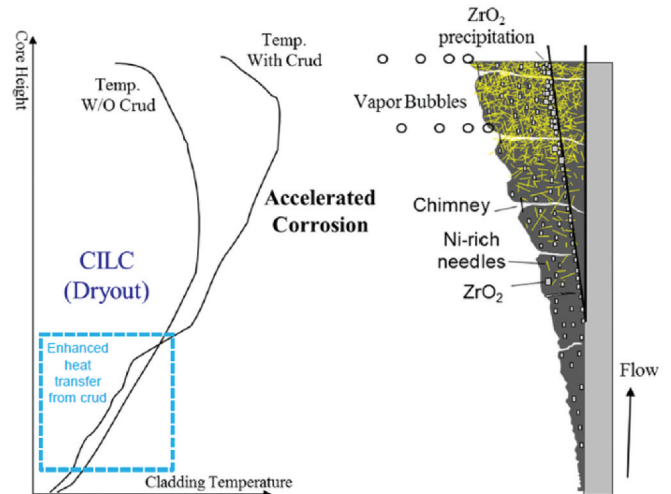


Fig. 7. Illustration of crud and CILC.

currently utilized manually controlled benchmarking process requires iteration on various inputs to a solution. The process requires a user to adjust a multiplier on crud source terms to match the corresponding threshold limit with the predicted BOA core boron. This value represents observed or measured CIPS for the plant. The process, as noted, is iterative, sometimes requiring several runs to narrow on a final crud source term multiplier for a given plant cycle to match the observed plant risk or data.

The BOA-predicted CIPS results are often conservative for non-CIPS cycles since the cycle is typically benchmarked up to the CIPS threshold (i.e., no observed core-wide CIPS, and so it is just under a defined threshold value where CIPS would be observed, and therefore, it is unknown how much white space remains prior to CIPS occurrence). Making use of additional measured plant data via local flux maps taken with movable in-core detectors is a means for capturing local indications of CIPS prior to the observed global CIPS event.

To capture the potential flux responses within the risk analyses, the CIPS risk predictions calculated with the coupled ANC/FUELDUTYDRV/VIPRE-W/BOA codes can potentially assess the resulting nodal boron feedback on the neutronics-generated axial power, which can be directly compared to the measured assembly flux maps (local instrumentation measured AO). The focus of this modeling approach is a plant with movable in-core detectors that has sufficient nodal points between the grids to characterize any flux depressions attributed to boron within the crud deposit. The results have been promising based on initial proof of concept, but have yet to be implemented for plant applications.

The use of multiple coupled codes results in a longer run time and a more complex calculation process to link with the measured assembly flux maps in order to benchmark a plant to actual operation rather than a BOA generic threshold value based on a data fit of previous CIPS events. However, a wealth of information is available with a plant's flux maps that could expand the experience base beyond just the CIPS events to aid in benchmarking plant predictions. This would allow for a potential benefit in quantifying the risk of a plant being below a generic BOA threshold value if CIPS has not been experienced. There is utility interest in the quantification of the margin to a CIPS event for each reload cycle and the potential to take advantage of that white space.

The application of AI/ML to enhance the prediction of CIPS/CILC could potentially reduce the cost associated with crud mitigation via manual iterations and the quantification of margin to the threshold limit, which would allow for more aggressive fuel management, reduce or eliminate fuel cleaning, and allow for longer cycle lengths or reduced costs associated with fuel performance validation for various fuel designs. This method would allow for more rapid assessment of proposed fuel product changes and their effect on subcooled nucleate boiling, the boiling surface area, and the resulting crud deposition.

The ML model could create an easy-to-use surrogate model that would enable the rapid assessment of risk to CIPS/CILC and expand the user basis. Exploratory analysis was performed to identify ways in which ML could be utilized to improve CIPS risk assessments within the current engineering process. Two approaches were identified: (1) ML applications for CIPS with the standalone BOA code and (2) ML applications for CIPS calibration with the coupled ANC/FUELDUTYDRV/VIPRE-W/BOA code.

This work presents the exploratory analysis, performed specifically for the first approach, to identify ways in which ML could be utilized to create a surrogate model of the standalone BOA computer code to predict boron mass in the reactor core of selected nuclear power plants. Following the exploration analysis, the surrogate model will be tested against various plants and cycles for predictive capabilities compared to actual cycle operation for validating the model.

III.B. ML Models and Methods

III.B.1. BOA Model

The BOA risk assessment tool combines corrosion, physical chemistry, and thermal-hydraulic models to

predict the release of corrosion products (crud) from the out-of-core surfaces of a PWR, as well as deposition on the fuel and out-of-core surfaces. Crud deposits preferentially on upper fuel surfaces that have undergone subcooled nucleate boiling, and the boron within the coolant can concentrate within the boiling crud matrix via several potential mechanisms, with lithium borate precipitation being the favored mechanism for this model.

The BOA code was originally released in 2003. The code was developed at the request of utilities to provide the PWR industry with a CIPS risk assessment tool. CIPS (formerly, axial offset anomaly or AOA) had first appeared in the early 1990s with sufficient frequency in various plants to warrant a mitigation strategy to assess and understand the effects of system and core design changes on power asymmetry and the shutdown margin. The code includes many of the elements known to affect PWR corrosion product transport and boron deposition, with many models being empirical. These models have been further developed and modified according to accrued plant data, with improved modeling of chemistry with laboratory test loop and plant data. Subsequent versions of the code, v2.0, v3.0, v3.1, and v4.0 have been released.

III.B.2. Data Generation

A sufficient amount of data generated by BOA simulations were required to train the ML model. The data generation process used a Monte Carlo-type simulation, i.e., independently sampling input parameters to generate output results and to develop the data set between the inputs and the outputs.

The nuclear power plants selected for the ML study were two higher-duty Westinghouse PWRs. Both plants had concerns about CIPS/CILC risk, and the BOA input models were available. A total of 14 fuel cycles were analyzed to cover a wide range of conditions in these two plants. Based on current CIPS work, cycles q, r, and s of plant A and the cycles k, l, and m to v of plant B were selected for the ML study.

The base BOA input files were modified to facilitate the Monte Carlo simulation that is a feature of the automated sensitivity calculation within the BOA code. The feature allows users to perform a series of BOA calculations automatically. The automated sensitivity calculation randomly samples parameters within the user's specified range according to either a uniform or normal distribution.

For this study, all parameters were sampled according to a uniform distribution since the normal

distribution led to questionable output results in some trial runs. Although the automated sensitivity calculation allows the BOA input to sample up to 74 parameters per run, only some of the input parameters were considered to be important in predicting the crud boron mass in the core. Based on previous sensitivity studies, 15 parameters were selected and used. In total, approximately 155 000 runs of the BOA code version 4.0, for a total of 14 fuel cycles, were completed to generate training and validation data. As indicated in the [Sec. III.A](#), the maximum total core boron mass calculated by BOA for each of these runs was selected as the target variable for the ML-based surrogate.

III.B.3. Data Cleansing and Feature Engineering

Once the raw data had been generated with BOA, they were processed and prepared for ML model training. The main data cleansing operation performed was the identification and removal of nonconverged BOA runs. There was a small fraction ($<10\%$) of the total runs that did not converge due to the uniform sampling of model inputs; each of these runs were dropped from consideration before training the ML model.

In addition to parameters that were randomly sampled during the database generation, new model input features were defined by averaging the assumed power distribution axially, radially, and/or over depletion steps for the given cycle. Several options for this averaging were tested, and it was determined that defining the power distribution using three axial segments averaged over all radial channels and depletion time steps (effectively representing average values for the bottom, middle, and top portions of the fueled region within the core) offered the best balance between model complexity and performance. This technique for averaging the power distribution was judged to make physical sense as well because (1) crud is typically distributed on hotter regions toward the axial center of the core and (2) the CIPS phenomenon is the result of differences in power at the top and bottom of the core.

III.B.4. ML Model Generation

Before training the ML model, the following standard data science practices were employed:

1. The data set of all converged Monte Carlo runs (145 366 total) was broken up into training and test data sets using an 80/20 split. The use of a train/test split in

this manner allow for the model to be appropriately evaluated on test data that it has never seen, helping to prevent overfitting of the model. Note that this split was stratified over each unique plant/cycle to ensure that potential differences between power distributions were appropriately accounted for during training.

2. The set predefined input features were normalized using a standard scaler, effectively converting each input feature into the number of standard deviations away from the mean of its respective distribution defined by the training data. This prevented the model from potentially only considering input features with large absolute magnitudes (e.g., axially averaged heat flux) and ignoring those with small absolute magnitudes (e.g., rate of particle release from boiling regions).

Several modeling techniques were evaluated in this exploratory analysis, including ridge regression, gradient-boosted decision trees, and neural networks. While ridge regression and gradient-boosted trees were found to be able to adequately predict a total boron mass for a single cycle, neural networks were determined to be capable of making accurate predictions over all cycles. Because it was desirable to have a single model for making predictions versus several models trained to individual cycles, neural networks were selected as the final modeling technique for this work effort.

While a fully connected neural network architecture was assumed for all training, a combination of Bayesian optimization and k -fold cross validation was utilized to define a process for tuning the model hyperparameters. Specifically, fivefold cross validation was utilized to assess the performance of a model for a given set of potential neural network hyperparameters that describe the model architecture (number of nodes, number of layers, activation function, etc.) and parameters (weight initialization, optimizer, batch size, etc.). The k -fold cross-validation process involved splitting the training data set into k equal-sized folds, using $k-1$ folds for training and the remaining fold for validation. This process is repeated k times, with each fold being used once as the validation data.

The performance scores from each iteration were then averaged to provide a more reliable estimate of the final model's performance. Once model performance had been defined for a given set of hyperparameters, a Bayesian optimization process was employed to provide information on the next best set of hyperparameters to consider based on the performance of previous tests (while balancing exploration of new areas within the hyperparameter space versus exploitation of previously tested areas exhibiting

good performance). This process was repeated 224 times to identify an optimal set of hyperparameters for the model.

III.C. Results and Discussions

Model performance was first assessed through a comparison of ML model-predicted total boron mass and the target BOA-predicted total boron mass for all points within the test data set, which were completely unseen by the model during training. Figure 8 illustrates a scatterplot of the BOA-predicted versus the ML-predicted total boron mass for the test data. Figure 9 provides an alternate view of model performance by plotting the percentage of test data points that fell within a given residual (defined as ML model-predicted – BOA-predicted total boron mass). This illustrates that a large fraction of test data were predicted with small residuals; for this model, approximately 95% of the test data were predicted within ± 0.035 lbm of the BOA result, and 100% of the test data were predicted within ± 0.20 lbm of the BOA result. This model performance was judged to be acceptable given that the BOA manual indicated that the benchmarking process or calibration of inputs should be within ± 0.10 lbm of the estimated boron mass obtained from the correlation between boron mass and the CIPS deviation.

In addition to assessing model predictions of the artificially generated test data, it was also desirable to determine how the model performed in the prediction of previously calibrated explicit BOA runs. To do this, inputs from the standalone BOA runs of plant A and plant B were obtained and put into a format that was readable by the neural network model. Table II illustrates the ML model-predicted

total boron mass versus the BOA total boron mass obtained from the respective output files. For each of the plant A and plant B cycles, the ML model was able to accurately predict total boron mass with a maximum residual of ~ 0.02 lbm. This level of error agreed with that observed when making predictions of the artificially generated test data set described previously and indicated that the model was operating as intended.

The ML surrogate model was also found to have significant benefits in execution time relative to the base BOA code. For example, to predict maximum core mass for a single cycle, the BOA execution time was around 120 s, while it took the ML surrogate model 0.03 s, representing a speed increase of four thousand times. This improvement to execution time can be leveraged as part of a calibration process that determines the input(s) that satisfies an observed (or some other desired target) core boron mass.

Figure 10 illustrates the application of the ML surrogate model for this use case, where the trajectory represents repeated modifications to a single model input (steam generator crud source term) to obtain a target core boron mass of 0.38 lbm. The calibration process was difficult and time consuming when performed manually with the BOA code, but with the ML surrogate model, it was achieved in less than 5 s.

III.D. Summary of ML Model Development for CIPS Risk Assessment

This study provided a demonstration of ML applications for BOA CIPS/CILC analysis. The work was

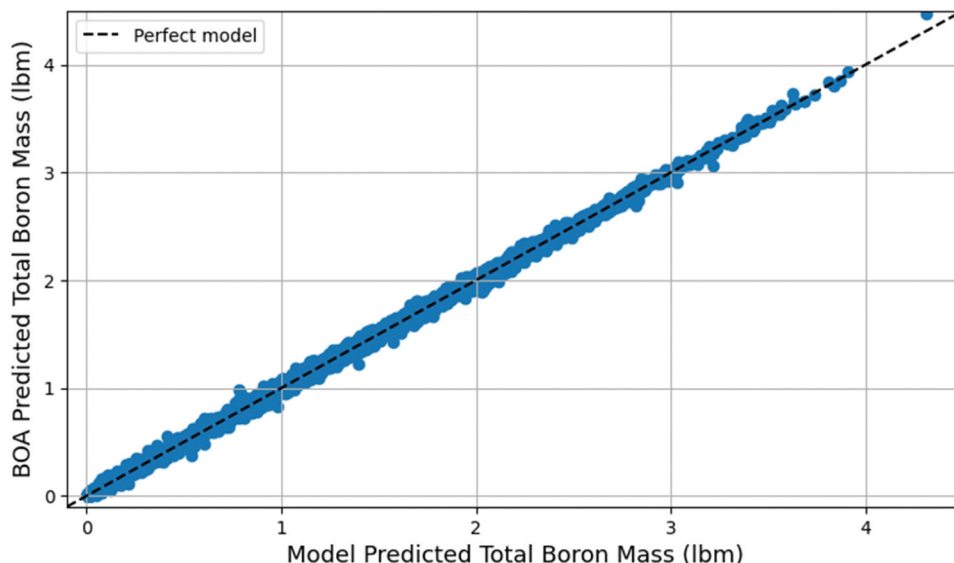


Fig. 8. Results of best fit neural network: BOA (observed) versus ML surrogate model (predicted) boron mass.

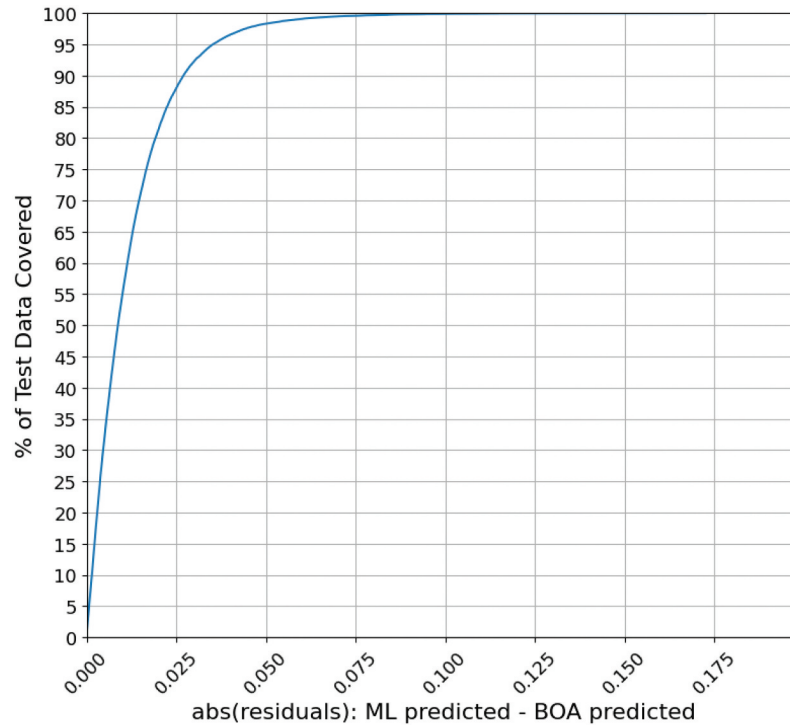


Fig. 9. Results of best fit neural network: fraction of test data versus residuals.

focused on the generation of a ML surrogate model to be utilized as an aid in the engineering process for determining CIPS risk using the BOA code. The trained ML surrogate model was successful in demonstrating that the ML model could make accurate predictions of the maximum total core boron mass across different cycles and PWR plants. It was judged that the goal of utilizing the ML model for the prediction of CIPS is feasible.

The ML surrogate model runs at a fraction of the computational cost (milliseconds) compared to the BOA

computer code, and enables applications requiring a high volume of runs, such as design optimization, margin quantification, and fuel cost reduction, that would otherwise be more time consuming using the computationally intensive code. The potential benefits of the ML surrogate model include the establishment of baseline risk for assessing changes in plant operation, use as a rapid screening tool for assessing fuel management strategies, and providing an easy-to-use model for expanded user basis.

TABLE II

ML Model Predictions Versus Previous
Plants A and B BOA Runs

	ML- Predicted Total Boron Mass (lbm)	BOA- Predicted Total Boron Mass (lbm)	Residual (lbm)
Plant A cycle q	0.3367	0.3259	0.0108
Plant A cycle r	0.2900	0.2803	0.0097
Plant A cycle s	0.2250	0.2258	-0.0008
Plant B cycle k	0.3678	0.3831	-0.0153
Plant B cycle l	0.5381	0.5199	0.0182
Plant B cycle m	0.7242	0.7145	0.0097

IV. CONCLUSIONS

Developments of AFQ approaches allow for opportunities to speed up the assessment and validation of new fuel designs with respect to performance and safety metrics. AFQ allows for shortening the time to develop and qualify new fuel for higher plant performance, in addition to reducing development costs through improved efficiency. Westinghouse utilizes advanced M&S technologies as part of their integrated and comprehensive AFQ vision through improved and accelerated fuel performance predictions.

This paper provided two example applications, prioritized by Westinghouse, that use ML technology as part of their advanced M&S suite for fuel thermal-hydraulic applications, including methodologies under development

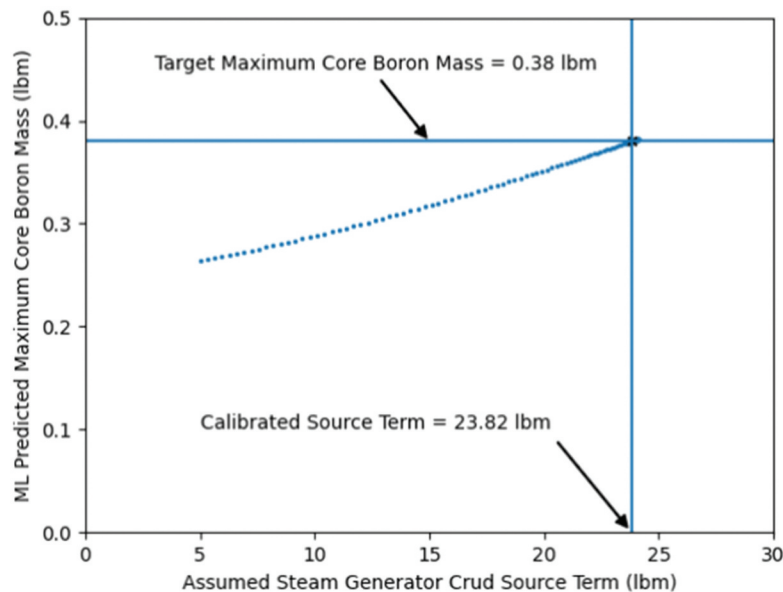


Fig. 10. Example of an input calibration use case of the ML surrogate model.

for the prediction of CHF for fuel thermal margin assessment and surrogate model development for CIPS risk prediction to enhance fuel operational performance. The CHF correlation development using ML technology for PWR fuel provides improved prediction performance over the traditional/empirical approach with improved margins to the DNBR limit, in addition to speeding up the schedule of correlation development for future fuel designs considerably in support of AFQ. ML technology implementation was also demonstrated in developing a surrogate model for CIPS risk prediction, allowing for similar prediction capability to the original approach, but with significantly reduced computational time and cost, again supporting AFQ.

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No potential conflict of interest was reported by the authors.

Author Contributions

CRedit: **Emre Tatli:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation,

Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing; **Yixing Sung:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Alex Mace:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Jun Liao:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing; **Jesse Fisher:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing; **James Spring:** Writing – original draft, Writing – review & editing; **Zeses Karoutas:** Conceptualization, Writing – review & editing; **Scott Sidener:** Conceptualization, Writing – review & editing.

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